

Hitesh Pattanayak

Senior Software Engineer | Backend Systems, Data Engineering & AI

[✉ pattanayak.hitesh03@gmail.com](mailto:pattanayak.hitesh03@gmail.com) [☎ +91 9503955631](tel:+919503955631) [📍 Bhubaneswar, Odisha, India](#) [🔗 GitHub](#) | [🌐 LinkedIn](#) [🌐 Portfolio](#)

Professional Summary

Senior Software Engineer with 10+ years of experience building production-grade backend systems, data pipelines, and AI-integrated services across cloud-native and enterprise environments. Recent work bridges deep backend expertise with applied AI engineering – designing RAG pipelines, building LLM-powered developer tools, and processing petabyte-scale M365 data for an enterprise data protection platform. Core stack: Go, Python, Kubernetes, Azure.

Technical Skills

- **Languages:** Go, Python, TypeScript
- **Data Engineering:** Databricks, Apache Spark, Delta Lake
- **Cloud & Infrastructure:** Kubernetes, Docker, Azure, AWS
- **Databases:** CosmosDB (vector + relational), PostgreSQL, Elasticsearch
- **Protocols & APIs:** gRPC, REST, GraphQL, OpenAPI
- **Monitoring & Observability:** OpenTelemetry, Kibana, Elastic Stack
- **AI & LLM Integration:** Azure OpenAI, Anthropic Claude API, RAG Pipelines, Vector Search, Embeddings, Few-Shot Prompting

Professional Experience

Improving (formerly Infracloud Technologies)

August 2022 - Present

*Infracloud Technologies was acquired by Improving in May 2025.
Roles and client engagements below span both.*

Senior Data Engineer | M365 Data Intelligence Platform

July 2025 - Present

- Architected end-to-end data pipeline infrastructure processing petabyte-scale M365 workload events using Databricks, Apache Spark, and Delta Lake
- Implemented high-volume streaming data processing with Azure Event Hubs and Container Apps
- Built Data Ingestion and Discovery services in Go, enabling real-time analytics and AI-enhanced search over M365 backup data
- Designed RAG pipeline for semantic search over M365 backup data – CosmosDB hybrid vector search, Azure OpenAI embeddings, and natural language to metadata filter conversion via few-shot Chat Completions
- Built Elastic dashboard changelog tool: diffs unreadable `.ndjson` files and generates human-readable changelogs via Anthropic API
- Built AI-assisted security fix tool: reads Cycode security findings and applies targeted, context-aware code fixes using LLMs
- Built On-call Copilot – a Claude Code plugin system covering 30+ alert types across data pipeline services; an engineer pastes an alert URL and gets root cause analysis with remediation in under 2 minutes; achieved 100% team adoption

Senior Software Engineer | Enterprise Authentication & User Management

November 2024 - June 2025

- Developed user management microservice handling authentication and authorization for an enterprise multi-tenant SaaS platform
- Built REST APIs in Go (Chi + OpenAPI) serving thousands of users across multiple organizations
- Integrated Auth0 with custom RBAC supporting 100+ permissions across Azure and Kubernetes workloads
- Designed CosmosDB data layer with optimized partitioning strategies for query performance at scale

Software Engineer | M365 Data Protection SaaS (Corso)

January 2024 - October 2024

- Engineered backend systems for a commercial SaaS control plane built on the open-source Corso engine, serving enterprise customers with M365 backup, restore, and export
- Developed CLI tooling and backend services for Exchange, OneDrive, SharePoint, and Teams data operations
- Implemented Microsoft Graph API integration and multi-backend storage support (S3, filesystem)
- Built CI/CD pipeline and comprehensive test coverage for a production SaaS product with paying enterprise customers

Software Engineer | Kubernetes Infrastructure & Platform Engineering

August 2022 - December 2023

- Built bare metal Kubernetes cluster provisioning platform with custom controllers and reconciliation logic
- Developed bootstrap service for private client networks with centralized SaaS integration
- Implemented bare metal provisioning using DHCP, TFTP, and Tinkerbell framework
- Designed agent-based architecture for distributed bare metal cluster management

Senior Consultant | Thoughtworks

October 2021 - July 2022

High-Performance Financial Trading Platform

- Architected microservices-based cryptocurrency trading platform for Voyager Inc.
 - Engineered time-series data management with TimescaleDB and PostgreSQL for sub-millisecond query performance
 - Built event-driven architecture with Apache Kafka for real-time market data and order processing
 - Deployed production-grade observability with Datadog, achieving 99.9% uptime
-

Software Engineer | Sureify

August 2020 - September 2021

Visual Platform Development

- Developed visual configuration platform for insurance workflows with drag-and-drop functionality
 - Built responsive React frontend and Node.js backend APIs for workflow management
 - Implemented microservices architecture for scalability and enhanced user experience
-

Software Developer | Cognizant & Infosys

January 2018 - July 2020

- Implemented RPA solutions and web applications for business process optimization
 - Built automated workflows for legacy enterprise system integration
-

Education

Bachelor of Technology in Electrical & Electronics Engineering

Gandhi Institute for Technological Advancement (GITA) Biju Patnaik University of Technology (BPUT), Bhubaneswar *March 2015*

Publications & Speaking

Books

- "Ultimate Certified Kubernetes Application Developer (CKAD) Certification Guide" - [OrangeAva](#)
- "Modern API Design with gRPC" - [OrangeAva](#)

Conference Talks

- "gRPC Load Balancing" - GopherCon 2023 - [YouTube](#)

- "Microservice Communication using gRPC" - [LinkedIn](#)

Technical Blogs

- "Bare metal K8s cluster provisioning using AWS EKSA" - [InfraCloud](#)
 - "Understanding gRPC Concepts, Use Cases & Best Practices" - [InfraCloud](#)
-

Personal AI Projects

- **Status Updater:** Scans JIRA, Confluence, and GitHub repos to auto-generate weekly work summaries
 - **Blog Generator:** AI-powered blog generation workflow (Claude/OpenAI) from structured idea files
 - **AI Chat Assistant:** RAG-based conversational assistant on personal blog using TF-IDF retrieval, Netlify Functions, and OpenAI gpt-4o-mini
-

Open Source Contributions

- [AWS EKS Anywhere](#) - Kubernetes cluster lifecycle management
 - [Corso](#) - Microsoft 365 data protection
 - [GPTScript AI](#) - AI automation framework
-

Recognition & Community

- **AWS Community Builder** (2023-2024) - Recognition for technical contributions and community engagement